Turning the tables: Exploring implicit causality effects on cataphoric pronouns

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Background: Cataphoric pronouns (where the pronoun occurs before the antecedent) provide a valuable window into dependency construction during comprehension. Prior work shows that anaphoric pronouns trigger an active search for an antecedent [1,2]. Clearly cataphoric pronouns initiate predictive processing, with comprehenders actively anticipating a potential antecedent to resolve the cataphoric pronoun as early as possible [3, 4].

A separate line of research investigates how pronoun resolution is guided by semantic and discourse-level information. A large body of work shows that verb semantics, particularly implicit causality (IC), guides the interpretation of anaphors (e.g. [5]). It is well known that in explanation configurations (with the connective *because*), **stimulus-experiencer** verbs (e.g. *to fascinate, to upset*) elicit an expectation that a subsequent pronoun refers to the preceding **subject**; hence, these are called **IC1** verbs. In contrast, **experiencer-stimulus** verbs (e.g. *to treasure, to resent*) bias reference to the preceding **object** (**IC2**). IC effects are observed crosslinguistically and are robust in online processing [6] as well as in offline interpretation [7, 8]. A finding that has received less attention is that IC verbs' **Emotional Valence** (**EV**) also guides anaphoric pronoun interpretation: *negative* events are more likely to be attributed to the preceding subject, while *positive* events more likely attributed to the preceding object [7].

Crucially, IC research effects have focused on anaphoric pronouns, i.e. pronouns following the verb. A study testing how cataphoric interpretation is influenced by the verb IC would shed light on the tension between (a) the forward-looking 'impatient' active search mechanism that seeks to link a cataphoric pronoun with an antecedent as early as possible [3,4], thus favoring the subject and (b) subsequent information from IC verb semantics, encountered *after* the pronoun. How does incremental context update modulate active search and interpretation? Do IC effects persist especially with IC2 verbs, even when the pronoun precedes the verb? Does passive voice alter the bias to the stimulus in IC contexts? To the best of our knowledge, there are no studies that systematically test if cataphoric pronoun interpretation is influenced by verb IC.

Experiment: We investigate this question with two sentence completion experiments in **English** that test whether and how strongly cataphoric pronoun configurations show IC verb effects. Exp1 tested active voice, and Exp2 tested active and passive voice (see Table 1).

Method and design: Exp1-2 were sentence completion tasks implemented in Qualtrics. Participants read sentences (24 targets, 36 fillers) and, in a variant of the typical sentence completion task, filled in blanks. On target trials, these blanks were in the *because*-clause preceding a cataphoric pronoun (Table 1). Target trials consisted of an initial *because*-clause (with a blank) followed by a main clause with two same-gender proper names as the subject and object of an IC verb. In Expt 1 (n=44), we manipulated verbs' **IC bias** (IC 1, IC2) and **emotional valence** (EV, positive or negative). In Expt 2 (n=35), we manipulated **IC bias**, **emotional valence** and voice (Active or Passive).

Results. **Expt 1** shows IC effects (glmer, $p<0.001$): pronouns are interpreted in line with the verb's IC bias (Fig1). *But* when a sentence contains an **IC2** verb with negative EV, the IC bias is significantly weaker ($p=0.02$, rightmost **bar**): pronouns are relatively more likely to receive a **subject** interpretation (interpreted as referring to the *first-mentioned* referent, against IC bias). Thus, when the pressure to resolve a dependency as early as possible converges with effects of EV, this weakens the IC effects of the following verb. **Expt 2** confirms that IC effects persist with cataphora (Fig 2): overall, IC1 verbs elicit more **subject** interpretations and IC2 verbs more **object** interpretations. Crucially, IC2 verbs with negative EV again show **weakened** IC effects ($p=0.01$), especially in active voice, like Expt 1. **In sum,** we show that IC effects persist even in 'active search' contexts with cataphora. However, when the pronoun precedes its antecedent (i.e. the parser engages in an 'impatient' search and tries to identify the antecedent as early as possible) this active search is strong enough to impact offline pronoun interpretation when it aligns with

emotional valence effects: We find higher rates of IC-incompatible subject choices with negative EV IC2 in both studies. Future work will test how these effects emerge in real-time processing.

Table 1: Example items (**Exp 1** tested only active voice, **Exp 2** tested both active and passive)

	Verb's implicit causality favors **subject** (IC1)	Verb's implicit causality favors the **object** (IC2)
Active	Because he _____, **Jack**$_{STIM}$ {fascinated$_{POS}$/upset$_{NEG}$} **Sean**$_{EXP}$.	Because he _____, **Jack**$_{EXP}$ {treasured$_{POS}$/resented$_{NEG}$} **Sean**$_{STIM}$.
Passive	Because he _____, **Sean**$_{EXP}$ was {fascinated$_{POS}$/upset$_{NEG}$} by **Jack**$_{STIM}$.	Because he _____, **Sean**$_{STIM}$ was {treasured$_{POS}$/resented$_{NEG}$} by **Jack**$_{EXP}$.
***Instructions**: Fill in the blanks with the most natural continuation that comes to mind.*		

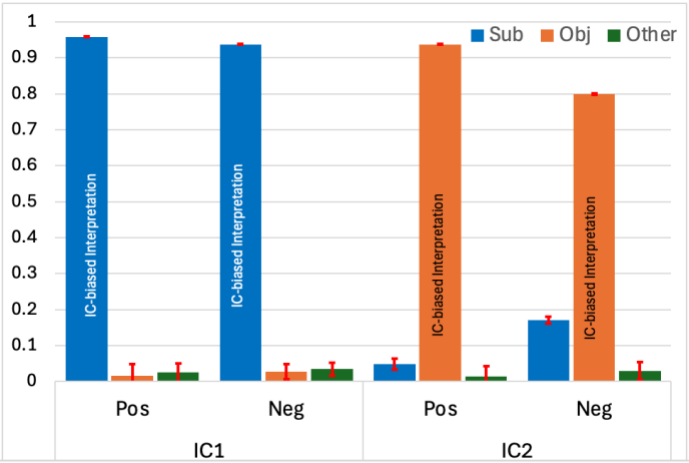

Fig.1: Expt 1: How often do people interpret the cataphoric pronoun as referring to the subsequent **subject** or **object**? [active voice]

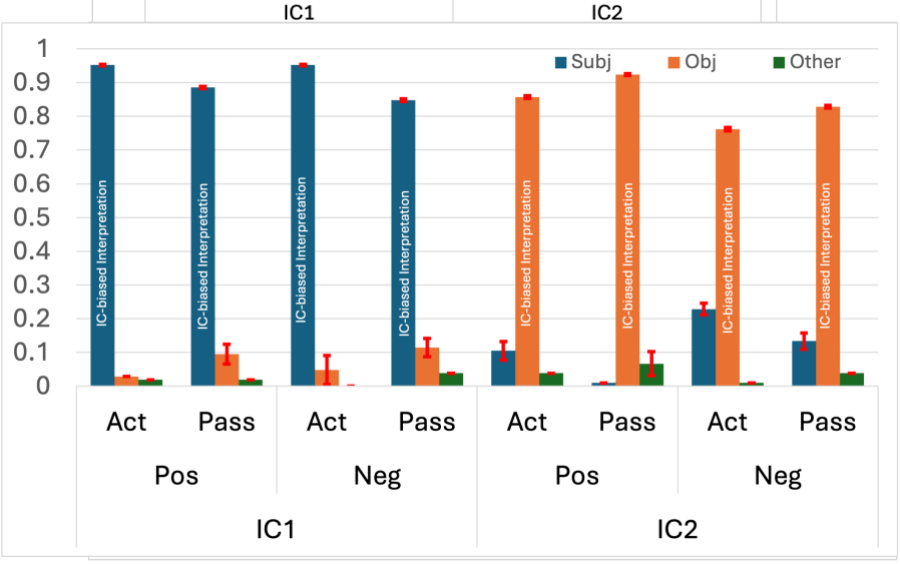

Fig.2: Expt 2: How often do people interpret the cataphoric pronoun as referring to the subsequent **subject** or **object**? [active and passive voice]

Note: In passives, the referent in by-phrase (e.g. **Jack**) is still labelled as the '**subject**' in Fig.2 (see Table 1), to facilitate comparison with active voice (whatever thematic role is the **subject** in active voice is in the **by-phrase** in the passive voice)

References: [1] Badecker & Straub '02. *JEP*. [2] Sturt '03 *JML*. [3] van Gompel & Liversedge, '03 *JEP:LMC*. [4] Fedele & Kaiser '14 *PLC.* [5] Garvey and Caramazza '74 *LI*. [6] Hartshorne '14. LCN [7] Ferstl et al. '11 *Beh.Res.Meth*. [8] Hartshorne & Snedeker '13. *LCP*.